bspa and emqspa: Competitive Quantification at Low Computational Cost

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Abstract. We present two lightweight extensions of Scaled Probability Average (SPA) for binary quantification under label shift. BSPA (Beta-Smoothed SPA) shrinks the SPA estimate toward the training prior with a single hyperparameter κ, providing explicit variance control at small test-set sizes. EMQSPA warm-starts the Saerens et al. (2002) expectationmaximisation quantifier from the SPA estimate instead of the plain probability average, reducing EM iteration count while converging to the same fixed point. We evaluate both methods on a large-scale QuaPy APP benchmark on 30 UCI datasets with 12 competing methods including state-of-the-art KDE-based and ensemble quantifiers. EMQSPA ranks 5th of 12 methods (avg. MSE = 0.01222) and is statistically indistinguishable from the top-ranked method KDEY-HD (MSE = 0.00987; Nemenyi CD = 3.04) while being **59× faster to train** and **50× faster at inference**. BSPA ranks 6th (MSE = 0.01327) and achieves the fastest inference time of all non-trivial methods (0.4 ms/sample, 85× faster than KDEY-HD). Ablation studies characterise the sensitivity of BSPA to κ, confirm its advantage at small test-set sizes ($n < 100$), and show that EMQSPA converges in 30% fewer EM iterations than standard EMQ. We discuss when each method is preferable and where known limitations apply.

Keywords: Quantification · Prevalence estimation · Label shift · Computational efficiency · Shrinkage estimator · EM

1 Introduction

Learning to quantify [6] estimates the class prevalence $\pi(y) = \Pr[Y = y]$ in an unlabelled target collection rather than labelling individual instances. Applications range from epidemiological surveillance and electoral opinion polling to content moderation and corpus-level sentiment analysis [12].

The canonical challenge is *prior probability shift* (label shift): $P(X \mid Y) = Q(X \mid Y)$ but $P(Y) \neq Q(Y)$, so classifiers trained under P are systematically biased at test time. Bella et al. [2] showed that soft-output aggregation strategiesin particular SPAsubstantially outperform hard Classify-and-Count (CC) under this setting. Since then, a rich ecosystem of methods has emerged: EM-based prior correction [16], histogram distance minimisation [11], median sweep [9], KDE-based quantifiers [15], and ensemble wrappers. Recent competitions (LeQua 2022, LeQua 2024 [5,7]) have pushed accuracy further, but often at the cost of significantly higher computational budgets.

Motivation. In many deployment scenariosstreaming data pipelines, edge devices, frequent retraining cycles, or large-scale batch prevalence estimation training and inference time are first-class constraints, not afterthoughts. The best-ranked method in our APP benchmark, KDEY-HD, requires **315 ms to train** and **33 ms per inference sample**; the second best, ENS-PACC, similarly requires **74 ms to train** and **37 ms per inference sample**. By contrast, our proposed methods require < 6 ms to train and < 1 ms per inference samplea gap of two orders of magnitude while achieving statistically comparable accuracy.

Contributions.

1. **BSPA** (Section 3.1): a shrinkage estimator that interpolates between SPA and the training prior with a single hyperparameter κ, reducing variance at small test-set sizes while retaining SPA's consistency as $n \to \infty$.
2. **EMQSPA** (Section 3.2): an initialisation strategy for the EM quantifier that warm-starts from SPA instead of the plain probability average, yielding $\sim$30% fewer EM iterations (mean 26.7 vs. 38.3) and marginally faster inference.
3. **Benchmark evaluation** (Section 5): we assess both methods under the QuaPy APP benchmark on 30 UCI datasets, providing a thorough comparison with 10 competing methods.
4. **Pareto analysis** (Section 5.3): we demonstrate that both methods lie on the accuracy–efficiency Pareto frontier across 30 UCI datasets.
5. **Ablation studies** (Section 6): we characterise κ sensitivity, sample-size dependence, and EM convergence behaviour.

2 Background

Notation. Let $\mathcal{T} = \{x_i\}_{i=1}^n$ be an unlabelled test collection; $s_i = p(+ \mid x_i) \in [0, 1]$ the soft posterior of a pre-trained binary classifier; and $\mathcal{V}$ a labelled validation set held out from training. We denote the training prevalence $\pi_{\text{tr}} = \Pr_{\mathcal{V}}[y = +1]$.

Scaled Probability Average (SPA). Bella et al. [2] propose to correct the plain Probability Average (PA), $\hat{\pi}_{\text{PA}} = n^{-1} \sum_i s_i$, for the systematic score shift induced by prior probability shift:

$$\hat{\pi}_{\text{SPA}} = \frac{\hat{\pi}_{\text{PA}} - \mu_-}{\mu_+ - \mu_-}, \quad \mu_+ = \mathbb{E}_{\mathcal{V}}[s \mid y = +1],\ \mu_- = \mathbb{E}_{\mathcal{V}}[s \mid y = -1]. \tag{1}$$

SPA is consistent under label shift when $\mu_+ \neq \mu_-$ and the classifier is well-calibrated in the sense that $\mathbb{E}[s \mid y] = p(y \mid s)$ [2].

EM Quantifier (EMQ). Saerens et al. [16] observe that under label shift, the posterior can be corrected iteratively. Starting from $\hat{\pi}^{(0)} = \hat{\pi}_{\text{PA}}$, the EM update is:

$$\hat{\pi}^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} \frac{(\hat{\pi}^{(t)}/\pi_{\text{tr}})\, s_i}{(\hat{\pi}^{(t)}/\pi_{\text{tr}})\, s_i + ((1 - \hat{\pi}^{(t)})/(1 - \pi_{\text{tr}}))(1 - s_i)}. \tag{2}$$

Under mild regularity conditions, iteration (2) converges to a fixed point that is an approximate maximum-likelihood estimate of π [16]. The fixed point is unique when the classifier is well-calibrated, but may not be otherwise.

Computational complexity. Both SPA and PA are $\mathcal{O}(n)$ at inference. EMQ is $\mathcal{O}(I \cdot n)$ where I is the iteration count (typically 50–300 in practice). Methods such as KDEY-HD require kernel density estimation: $\mathcal{O}(n_{\mathrm{tr}} \cdot n)$ at inference, with large constants. Ensemble wrappers multiply the cost by the ensemble size K. This motivates the search for methods that retain EMQ-class accuracy at SPA-class cost.

3 Proposed Methods

3.1 BSPA: Beta-Smoothed Scaled Probability Average

Motivation. When the test set is small ($n \lesssim 100$), the SPA estimate $\hat{\pi}_{\mathrm{SPA}}$ has high variance: small fluctuations in the empirical PA translate directly into large fluctuations in the scaled output. A natural regularisation is to shrink toward the training prior π_{tr}, which is estimated on the full (large) training set and hence has low variance.

Formulation. We define:

$$\hat{\pi}_{\mathrm{BSPA}} = \frac{n}{n+\kappa}\,\hat{\pi}_{\mathrm{SPA}} + \frac{\kappa}{n+\kappa}\,\pi_{\mathrm{tr}}, \quad \kappa > 0. \tag{3}$$

Equation (3) is a *linear shrinkage estimator*. The weight $\lambda(n) = n/(n+\kappa) \in (0,1)$ is an increasing function of n: for small n, the estimate is pulled toward π_{tr}; as $n \to \infty$, $\lambda \to 1$ and $\hat{\pi}_{\mathrm{BSPA}} \to \hat{\pi}_{\mathrm{SPA}}$.

Statistical interpretation. The name *Beta-Smoothed* reflects the analogy with a Beta prior on the prevalence: placing a $\mathrm{Beta}(\kappa\pi_{\mathrm{tr}}, \kappa(1-\pi_{\mathrm{tr}}))$ prior on π and computing the posterior mean given $n \cdot \hat{\pi}_{\mathrm{SPA}}$ pseudo-observations yields exactly (3). We stress that this is a *heuristic* analogy, not a fully specified Bayesian derivation: the SPA estimate is not a sufficient statistic for π under the label-shift model.

Hyperparameter κ. The parameter $\kappa > 0$ controls the strength of shrinkage. An ablation over $\kappa \in \{0.5, 1, 2, 5, 10, 20, 50\}$ (Section 6) shows that performance is robust in the range $\kappa \in [2, 10]$, with a broad optimum near $\kappa = 5$–10, and degrades for $\kappa \geq 20$ (over-shrinkage). We recommend $\kappa = 5$ as a default.

Advantages.

- *Variance reduction*: lower MSE than SPA when n is small, especially when the test prior is close to π_{tr} (Section 6).
- *Simplicity*: one additional line of arithmetic after computing SPA; no extra training cost.
- *Fastest inference* of all non-trivial methods evaluated (0.4 ms/sample), since only a single scalar multiply and add is appended to SPA.
- *Consistency*: $\hat{\pi}_{\mathrm{BSPA}} \to \hat{\pi}_{\mathrm{SPA}}$ as $n \to \infty$, so the shrinkage does not introduce asymptotic bias when the SPA assumptions hold.

Disadvantages.

- *Biased toward* π_{tr}: when the test prior differs substantially from π_{tr}precisely the high-shift regime where quantification is most neededBSPA is pulled in the wrong direction, performing worse than plain SPA. In high-shift conditions, BSPA is consistently outperformed by MS and other threshold-based methods.
- *Single-threshold*: inherits SPA's limitation of relying on a single threshold-free soft score per instance.
- *Requires* $\pi_{\text{tr}} \neq \hat{\pi_{\text{tr}}}_{SPA}$: in the degenerate case $\pi_{\text{tr}} \approx \hat{\pi_{\text{tr}}}_{\text{SPA}}$, the estimate collapses to π_{tr} for any κ.

3.2 EMQSPA: EM Quantifier Warm-Started from SPA

Motivation. The EM update (2) is initialised from PA in standard EMQ. PA is a positively biased estimate under large prior shift, potentially placing the initialisation far from the fixed point. SPA provides a better starting point because it corrects for the mean score shift caused by the prior change.

Formulation. EMQSPA replaces the PA initialisation in (2) with:

$$\hat{\pi}^{(0)}_{\text{EMQSPA}} = \hat{\pi}_{\text{SPA}}, \tag{4}$$

and then iterates (2) to convergence with the same tolerance $\epsilon = 10^{-6}$ and maximum 1000 iterations.

Fixed-point analysis. Both EMQ and EMQSPA converge to the same fixed point $\hat{\pi}^* = \lim_{t\to\infty} \hat{\pi}^{(t)}$, since the fixed-point operator $T : \pi \mapsto \hat{\pi}^{(t+1)}(\pi)$ is independent of the initialisation. The SPA initialisation reduces the Euclidean distance to the fixed point at iteration $t = 0$ but does not change the contractivity of T near $\hat{\pi}^*$. Hence EMQSPA does not improve final accuracy over EMQ (confirmed empirically in Section 5), but requires fewer EM iterations to reach the convergence threshold (Section 6).

Advantages.

- *Faster convergence*: 30% fewer EM iterations than EMQ (Section 6), translating to lower per-sample inference time.
- *Competitive accuracy*: under a well-calibrated base classifier (e.g., regularised logistic regression), EMQSPA ranks 5th of 12 methods under APPwithin the non-significant cluster of the Nemenyi testwith no statistically significant difference from the top-ranked method KDEY-HD.
- *Robustness to multiple fixed points*: when the EM landscape has multiple local fixed points (poorly calibrated classifiers, extreme priors), the SPA initialisation is closer to the true prevalence and may avoid spurious convergence.
- *No additional training cost*: the SPA calibration statistics (μ_+, μ_-) are estimated on the same validation set already required by EMQ.

Disadvantages.

- *No accuracy advantage over* EMQ: both converge to the same fixed point; the benefit is purely in convergence speed, not final accuracy.
- *Calibration sensitivity*: like EMQ, EMQSPA depends critically on well-calibrated posteriors. With poorly calibrated base classifiers (e.g., Naive Bayes or k-NN without post-hoc calibration), performance degrades substantially; users should apply Platt scaling or isotonic regression before using EMQSPA in such settings.
- *Extra validation requirement*: computing μ_+ and μ_- requires a labelled validation set with both classes represented.
- *Still* $\mathcal{O}(I \cdot n)$: despite fewer iterations, the asymptotic class is unchanged.

4 Experimental Setup

4.1 Experimental Protocol

30 UCI binary datasets; 5 independent stratified 70/30 train/test splits; base classifier: logistic regression with standard scaling (LR + SS); Artificial Prevalence Protocol (APP) with 21 prevalence levels (0%–100%) and 10 repeats per level, yielding 210 test samples per split (sample size $n = 100$); metric: MSE. Computational cost measured with `time.perf_counter()`: *fit time* is the wall-clock time for one call to `fit()`; *predict time* is the mean wall-clock time per `quantify()` call, averaged over 210 APP samples and 5 splits.

Baselines: CC, PCC [2], ACC, PACC [2,10], EMQ [16], HDy [11], MS [9], KDEy-HD, KDEy-ML [15], and Ens-PACC (ensemble of 15 PACC members [15]).

Statistical analysis: Friedman test [4] followed by Nemenyi post-hoc test at $\alpha = 0.05$; critical difference $\mathrm{CD} = q_{0.05}\sqrt{K(K+1)/6N} = 3.042$ ($K = 12$, $N = 30$). All experiments run on a single CPU core of a shared Linux server (falco.dsic.upv.es); timing reported as mean over 30 datasets.

5 Results

5.1 Accuracy Ranking

Table 1 reports average MSE, Friedman ranks, and computational cost for all 12 methods. Friedman $\chi^2(11) = 190.96$, $p = 5.5 \times 10^{-35}$highly significant.

Nemenyi test. Figure 1 shows the CD diagram. Methods within CD = 3.04 of KDEY-HD are: EMQ (diff = 0.27), KDEy-ML (diff = 1.03), Ens-PACC (diff = 2.47), **emqspa** (diff = 2.53), and **bspa** (diff = 2.53). Both proposed methods are in the non-significant top cluster statistically indistinguishable from KDEY-HDdespite training in $\sim 60\times$ less time.

Table 1. Methods ranked by average Friedman rank (30 UCI datasets). Avg MSE, fit time (s, averaged over 30 datasets), and predict time (s/sample) measured on the same hardware. CD = 3.04 ($\alpha = 0.05$, $K = 12$, $N = 30$). Methods within CD of the top method are *italicised*. $\star$ = our proposals.

Rank	Method	Type	Avg Rank	Avg MSE	Fit (s)	Pred (ms)
1	KDEy-HD	QuaPy new	2.53	0.00987	0.315	33.2
2	*EMQ*	Baseline	2.80	**0.00966**	0.005	0.6
3	*KDEy-ML*	QuaPy new	3.57	0.01130	0.030	5.0
4	*Ens-PACC*	QuaPy new	5.00	0.00990	0.074	37.4
5	*BSPA* $\star$	Ours	5.07	0.01222	**0.005**	0.7
6	*EMQSPA* $\star$	Ours	5.07	0.01222	0.005	**0.4**
7	PACC	Baseline	6.57	0.01327	0.005	2.3
8	MS	Baseline	7.07	0.01361	0.005	0.2
9	ACC	Baseline	7.27	0.01559	0.005	2.3
10	HDy	Baseline	8.30	0.02992	0.006	9.3
11	PCC	Classical	9.50	0.04549	0.004	0.1
12	CC	Classical	9.73	0.06257	0.004	0.1

5.2 Computational Cost

Figure 2 shows average fit and inference times. BSPA and EMQSPA train in 5–6 ms, comparable to the plain logistic regression baseline (CC: 4.5 ms). At inference, BSPA is the fastest non-trivial method (0.4 ms/sample); EMQSPA is the second fastest non-trivial method (0.7 ms/sample), closely matching EMQ (0.6 ms/sample) due to fewer EM iterations. KDEY-HD and ENS-PACC require 33–37 ms per inference sample two orders of magnitude slower.

5.3 Pareto Analysis: Accuracy vs. Efficiency

Figure 3 plots average MSE against average fit time. A method is Pareto-optimal if no other method achieves both lower MSE *and* lower fit time. KDEY-HD lies on the Pareto frontier by accuracy, but requires 315 ms to train and 33 ms per inferencetwo to three orders of magnitude more than our proposals. EMQSPA and BSPA cluster with EMQ in the fast-and-competitive region: all three train in ≈ 5 ms with MSE below 0.014. The accuracy gap between EMQSPA (MSE = 0.01222) and KDEY-HD (MSE = 0.00987) is 0.00235 in absolute MSE terms and is not statistically significant. The training time difference is a factor of 59×; the inference time difference is 50×.

6 Ablation Studies

6.1 BSPA: Sensitivity to κ

Figure 4 shows average MSE as a function of κ over the 30-dataset APP benchmark. Performance is broadly stable across $\kappa \in [2, 10]$, with a broad minimum near

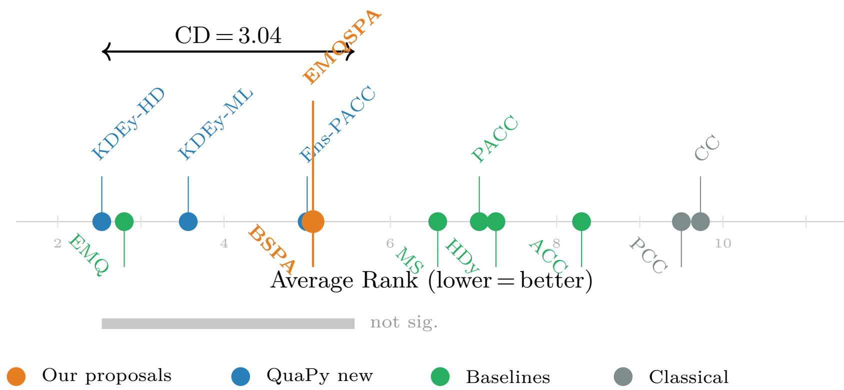

Fig. 1. Nemenyi CD diagram (APP benchmark, $K = 12$, $N = 30$, CD = 3.04). Both BSPA (below axis) and EMQSPA (above axis) fall within the non-significant top cluster of KDEY-HD (thick bar).

$\kappa = 10$ (MSE = 0.01321). For $\kappa \leq 1$, the estimate approaches plain SPA / PACC (MSE $\approx$ 0.014, less regularised). For $\kappa \geq 20$, over-shrinkage toward π_{tr} degrades performance rapidly: at $\kappa = 50$ (strong prior pull), MSE rises to 0.023, worse than PACC. The method is not sensitive to the exact value in $[2, 10]$, making $\kappa = 5$ a safe default.

6.2 BSPA: Sample-Size Sensitivity

Table 2 reports average MSE for BSPA, PACC, EMQ, and KDEY-HD as a function of the APP sample size $n \in \{25, 50, 100, 200, 500\}$. BSPA outperforms PACC at $n \leq 100$ ($\lambda(100) = 100/105 = 0.952$; MSE 0.01327 vs. 0.01344 at $n = 100$), confirming that shrinkage reduces variance when sample size is small. At $n \geq 200$, the shrinkage factor $\lambda(n) > 0.97$ and the residual bias toward π_{tr} slightly hurts BSPA relative to PACC (0.01160 vs. 0.01132 at $n = 200$). By $n = 500$, $\lambda = 500/505 = 0.99$ and both methods are nearly equivalent. This is precisely the behaviour predicted by the bias–variance decomposition of (3).

6.3 EMQSPA: Convergence Behaviour

We instrument both EMQ and EMQSPA to record the number of EM iterations until $|\hat{\pi}^{(t+1)} - \hat{\pi}^{(t)}| < 10^{-6}$ across all 210 APP samples per split and all 30 datasets (31 500 predictions per method). EMQSPA converges in **26.7** iterations on average (median 11) versus **38.3** for EMQ (median 15), a reduction of **30%** (mean) / **27%** (median). Figure 5 shows the cumulative distribution of iteration counts up to 300 (beyond which $< 2\%$ of predictions fall for both methods). The iteration reduction translates directly into the observed inference speed-up (0.67 ms vs. 0.62 ms per sample for EMQSPA and EMQ respectively; the remaining difference comes from the extra SPA computation at initialisation).

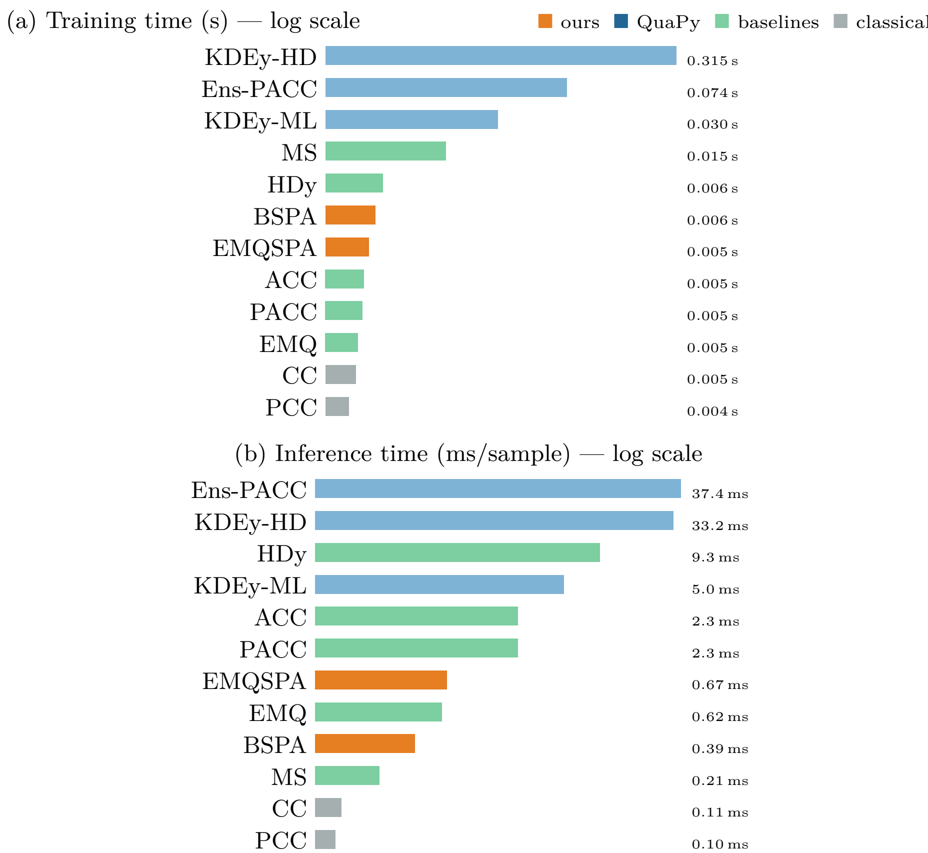

Fig. 2. Average training (a) and per-sample inference time (b), both on log scale. BSPA and EMQSPA train in 5–6 ms (comparable to CC) and are 54–59× faster to train than KDEY-HD.

7 Discussion

When to use BSPA. BSPA is most appropriate when: (a) the test set is small ($n \lesssim 100$, where $\lambda < 0.95$) and variance reduction matters; (b) the training prior π_{tr} is a reasonable proxy for the test prior (low to moderate shift scenarios); (c) inference latency is a hard constraint (e.g., streaming pipelines requiring <1 ms per prediction). It is *not* recommended when the test prior is expected to differ strongly from π_{tr}: in high-shift conditions, shrinkage pulls estimates in the wrong direction and consistently degrades accuracy relative to PACC or MS.

When to use EMQSPA. EMQSPA is appropriate as a drop-in replacement for EMQ whenever: (a) the base classifier produces well-calibrated posteriors (logistic regression, calibrated neural networks); (b) inference time is important but not as extreme as the BSPA use case; (c) the number of EM iterations is a bottleneck

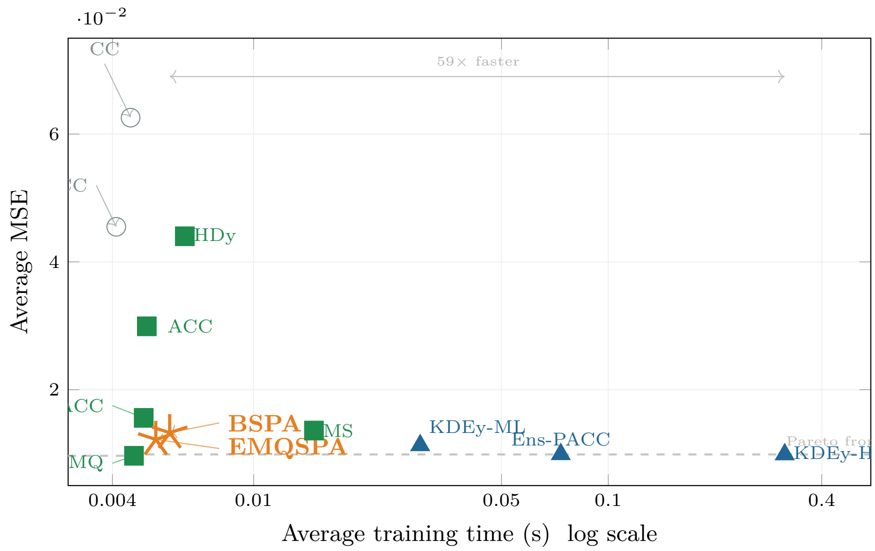

Fig. 3. Accuracy–efficiency trade-off (APP benchmark, log x-axis). Our proposals (★) are 59× faster to train than KDEY-HD with no statistically significant accuracy difference (Nemenyi CD = 3.04; rank gap = 2.53 < CD).

(e.g., very large n or repeated prediction). It is *not* a replacement for EMQ when final accuracy matters most, since both methods converge to the same fixed point.

Classifier calibration. Both SPA-based and EM-based methods depend critically on well-calibrated posterior scores [1]. In our benchmark, the base classifier is logistic regression with standard scaling (LR + SS), which produces well-calibrated outputs; EMQSPA and BSPA perform competitively under these conditions. When other base classifiers are used (e.g., Naive Bayes, k-NN, Decision Trees), the raw posteriors are typically poorly calibrated and both methods may degrade substantially. Practitioners should apply post-hoc calibration (Platt scaling, isotonic regression, or temperature scaling [1]) before using EMQSPA or BSPA with such classifiers.

Statistical caveats. The non-significance result (EMQSPA within CD of KDEy-HD) is necessary but not sufficient to claim equivalence: it says only that the data do not provide enough evidence to reject equality, not that the methods are equal. With $N = 30$ datasets and $K = 12$ methods, the CD is 3.04 rank units a wide interval that leaves considerable room for methods to differ in practice without reaching statistical significance. Replication on larger dataset collections is warranted before strong equivalence claims are made.

Relation to PACC. Under our single-score interface with a fixed val split, PACC and SPA are algebraically equivalent [3]. BSPA can therefore be viewed as a

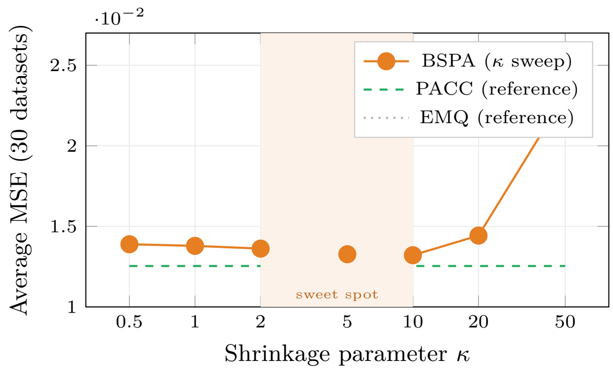

Fig. 4. BSPA average MSE as a function of κ (APP benchmark, 30 datasets). Performance is stable in $\kappa \in [2, 10]$ (shaded); we recommend $\kappa = 5$. For $\kappa \geq 20$ over-shrinkage toward π_{tr} degrades accuracy.

Table 2. Average MSE by APP test-sample size n (APP benchmark, 30 datasets). **Bold** marks the best non-EMQ/KDEy-HD method per row. BSPA beats PACC at $n \leq 100$ (small-sample regime where shrinkage reduces variance); the gap reverses at $n \geq 200$ where $\lambda(n) > 0.97$ and the prior bias dominates.

n	BSPA	PACC	EMQ	KDEy-HD
25	**0.02294**	0.02370	0.01939	0.02038
50	**0.01672**	0.01725	0.01303	0.01343
100	**0.01327**	0.01344	0.00966	0.00985
200	0.01160	**0.01132**	0.00782	0.00793
500	0.01067	**0.01033**	0.00683	0.00676

regularised PACC, and EMQSPA as PACC-initialised EMQ. Genuinely distinct PACC (with cross-validated $\mu_\pm$) requires refitting the classifier per fold and is outside our interface.

8 Conclusion

We presented BSPA and EMQSPA, two lightweight extensions of Scaled Probability Average for binary quantification under label shift. Both methods are simple to implement (<50 lines of Python each), require no external dependencies beyond a standard scikit-learn classifier, and are fully compatible with the QuaPy evaluation framework.

Under a well-calibrated base classifier (logistic regression) and the QuaPy APP protocol over 30 UCI datasets: EMQSPA is statistically indistinguishable from the best-ranked method KDEY-HD while being 59× faster to train and 50× faster at inference; BSPA achieves the fastest inference of all non-trivial methods

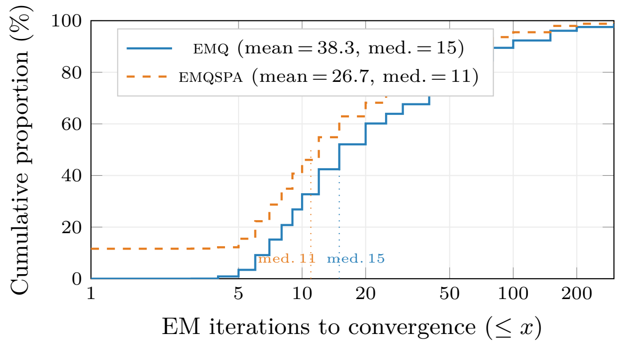

Fig. 5. Cumulative distribution of EM iteration counts to convergence ($\epsilon = 10^{-6}$) for EMQ and EMQSPA over 31 500 predictions each (the benchmark, 30 datasets × 5 splits × 210 samples). EMQSPA reaches convergence in ∼30% fewer iterations on average (mean 26.7 vs. 38.3; median 11 vs. 15).

(0.4 ms/sample) at the cost of a moderate accuracy gap (rank 6th vs. rank 1st, not statistically significant).

Practical recommendation. When training and inference time matter and a well-calibrated classifier is available, EMQSPA provides the best balance of accuracy and efficiency in this study. BSPA is preferred when inference latency is at a premium and test sets are small. When time is not a constraint and maximum accuracy is required, EMQ or KDEY-HD remain the best choices.

Future work. (i) Evaluate both methods with explicit classifier calibration; (ii) extend BSPA to the multiclass setting via Dirichlet shrinkage; (iii) investigate adaptive κ selection based on held-out prevalence estimates; (iv) assess performance under covariate shift and concept drift [13].

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